On the Decidability of $\mathcal{ALCI}_{RCC5}$: A Survey of the Problem, Its History, and a Proposed Solution

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Abstract

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with the five RCC5 base relations as roles, under composition-table semantics. Its decidability has been open since Wessel (2002/2003). We survey the history of the problem—from Cohn's original proposal (1993) through Wessel's systematic investigation and Lutz & Wolter's related undecidability results (2006)—and explain why all known undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$. We then present a proposed decision procedure: the *cover-tree tableau*, which decomposes models into PPI-oriented cover trees where only $\{\mathsf{DR}, \mathsf{PO}\}$ remain as open cross-edges. This decomposition, combined with the patchwork property of RCC5, yields finite blocking via rank-d signatures. The approach is supported by extensive computational evidence (911 test concepts, zero mismatches; 775/775 SAT concepts with cover-tree models) but has not been formally verified by human experts. We discuss why naïve tableau approaches fail for $\mathcal{ALCI}_{RCC5}$ and how the cover-tree tableau overcomes these obstacles.

Keywords

description logic, spatial reasoning, RCC5, decidability, tableau calculus, patchwork property

1. Introduction

The decidability of concept satisfiability in the spatially extended description logic $\mathcal{ALCI}_{RCC5}$ has been an open problem for over twenty years. This paper provides a self-contained overview of the problem, its intellectual history, the obstacles that have resisted solution, and a proposed decision procedure that—if correct—would settle the question positively.

The paper is organized as follows. Section 2 traces the history of $\mathcal{ALCI}_{RCC5}$ from Cohn's original 1993 proposal through Wessel's systematic investigation and Lutz & Wolter's topological undecidability results. Section 3 explains why standard undecidability reductions fail, including a concrete domino-encoding attempt that we verify is blocked. Section 4 discusses why naïve tableau approaches fail for complete-graph logics. Section 5 presents the split-forest model decomposition. Section 6 describes the cover-tree tableau and explains how it achieves blocking. Section 8 concludes with an assessment of what has been achieved and what remains to be done.

Disclaimer and AI-tool disclosure. This paper was prepared by Michael Wessel with significant assistance from generative-AI language tools: Claude (Anthropic, Opus 4.7) [41], GPT-5.4 Pro (OpenAI) [42], and GPT-5.5 (OpenAI) [16]. The AI tools were used to formalise the mathematical proofs, implement the cover-tree tableau decision procedure, conduct the computational verification campaigns, and draft the manuscript text including this overview. Wessel reviewed and edited the content as needed, directed the research, identified and corrected errors, and takes full responsibility for the publication's content—including any errors, mistakes, or misleading statements the AI tools may have introduced. See the Declaration on Generative AI at the end of the paper for further details. The results and proofs have **not been peer-reviewed or verified by human domain experts**; they are published as a discussion piece. The claims should not be taken as established results unless independently verified. Scrutiny, corrections, and feedback are invited.

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2. History of $\mathcal{ALCI}_{RCC5}$

2.1. Cohn's 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle \mathsf{PP} \rangle C$ for "some proper part satisfies C"—but Cohn left all decidability and complexity questions open.

2.2. Wessel's $\mathcal{ALCI}_{RCC}$ family (2002/2003)

Wessel [7, 8, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{RCC}$ *family*: description logics $\mathcal{ALCI}_{RCCk}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{RCCk}$ interpretation, every pair of domain elements is assigned exactly one RCCk base relation, and the composition table is enforced globally (*strong* EQ *semantics*: EQ is identity). Models are complete graphs with RCCk edge-colorings.

Wessel established the following:

- **Decidable cases:** $\mathcal{ALCI}_{RCC1}$, $\mathcal{ALCI}_{RCC2}$, and $\mathcal{ALCI}_{RCC3}$ are decidable, because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques.
- **Lower bounds:** $\mathcal{ALCI}_{RCC5}$ is PSPACE-hard; $\mathcal{ALCI}_{RCC8}$ is EXPTIME-hard.
- **Open:** The decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ was left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].
- $\mathcal{ALC}_{\mathcal{RA}_{SG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RA}_{SG}}$) makes it undecidable again.

$\mathcal{ALCI}_{RCC5}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RA}_{SG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$).

Remark 2.1 (Key difficulties)**.** $\mathcal{ALCI}_{RCC5}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 1. Reading: row = $S(b, c)$, column = $R(a, b)$, entry = possible relations for (a, c).
Key composition facts exploited in the cover-tree theory (Sections 5–6): $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ (DR rigidity); $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ (transitivity); and $\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ (PP can be *forced* by composition and universal filtering; see Example 6.2).

Table 1
The RCC5 composition table ($*$ = all five relations).

comp	DR(a, b)	PO(a, b)	EQ(a, b)	PPI(a, b)	PP(a, b)
DR(b, c)	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b, c)	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
EQ(b, c)	DR	PO	EQ	PPI	PP
PP(b, c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b, c)	DR	DR, PO, PPI	PPI	PPI	$*$

2.3. Wessel's grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{RCC8}$ [8], Wessel made two important discoveries:

1. Even-odd chains (Section 4.4.2). The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between $even$ and odd, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary n-bit counter, proving $\mathcal{ALCI}_{RCC8}$ is ExpTime-hard.

2. The grid coincidence obstruction (Section 4.4.3, Figure 11). Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

> "Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term."

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*.

These constructions—the even-odd chain, the grid encoding via TPP/NTPP, and the identification of the coincidence gap as the decisive obstacle—anticipated key technical ingredients of the later Lutz–Wolter proof by three to four years.

2.4. Lutz & Wolter (2006)

Lutz and Wolter [9] studied *modal logics of topological relations*, denoted L_{RCC8}, L_{RCC5}, etc. These have the same *syntax* as $\mathcal{ALCI}_{RCC}$ but are interpreted over *topological models*: the domain elements are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- L_{RCC8} is **undecidable** (domino tiling via functional TPP-chains, with the grid coincidence forced by the geometry of $\mathbb{R}^2$).
- $L_{RCC5}(RS^{\exists})$ is **undecidable** (reduction from $S5^3$ via supremum regions).
- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): "Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply."

The logic $\mathcal{ALCI}_{RCC5}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

Table 2
Why standard undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$.

Source Problem	**Technique**	**Missing in $\mathcal{ALCI}_{RCC5}$**
$\mathbb{N}\times\mathbb{N}$ domino	Graded modalities + transitivity	Number restrictions
$\mathcal{ALC}_{\mathcal{RA}}$ [6]	Post Correspondence Problem	Arbitrary role axioms
$\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ [3]	Non-empty CFG intersection	Non-disjoint roles
$\mathcal{ALCI}_{RCC8}$ (Wessel)	Grid via TPP/NTPP	TPP/NTPP distinction
L_{RCC8} (Lutz–Wolter)	Domino via topological TPP	Grid coincidence
$L_{RCC5}(RS^{\exists})$	$S5^3$ via supremum regions	Supremum closure

3. Why Undecidability Reductions Fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N}\times\mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{RCC5}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

3.1. Blocked reduction routes

Table 2 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

3.2. A concrete domino attempt in $\mathcal{ALCI}_{RCC5}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned} X \;=\; & \exists \mathsf{PPI}.\exists \mathsf{PPI}.C \;\sqcap\; \exists \mathsf{PPI}.\exists \mathsf{PPI}.D \\ & \sqcap\; \forall \mathsf{PPI}.\forall \mathsf{PPI}.(\forall \mathsf{PO}.(\neg C \sqcap \neg D) \sqcap \forall \mathsf{DR}.(\neg C \sqcap \neg D) \\ & \qquad\qquad\qquad \sqcap\; \forall \mathsf{PPI}.(\neg C \sqcap \neg D) \sqcap \forall \mathsf{PP}.(\neg C \sqcap \neg D)) \\ & \sqcap\; \forall \mathsf{PPI}.\exists \mathsf{PPI}.X \end{aligned}$$

The intent: each X-node has two PPI-grandchildren colored C and D (the "square"), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive $\forall\mathsf{PPI}.\exists\mathsf{PPI}.X$ forces infinite repetition.

One level is satisfiable. Without the recursive conjunct, X has a 3-element model: z PP y PP x, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* (EQ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, and the universals would require $z_2 \notin D$ (firing z_1's $\forall R.\neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$, and the universals are vacuously satisfied (the only neighbors of z are PP-predecessors, not PO/DR/PPI-neighbors).

Two levels is unsatisfiable. Adding $\forall\mathsf{PPI}.X$: the PPI-child y must itself satisfy X (without recursion), producing a new $C \sqcap D$-labeled element w as a PPI-child of z. But z is a PPI-grandchild of x, and x's universals fire: z's PPI-neighbor w must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \sqcap \forall\mathsf{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall\mathsf{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

Remark 3.1. The failure is not accidental. In $\mathcal{ALCI}_{RCC5}$, the only way to create "functional" successors is via PP/PPI chains, but $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{RCC8}$, the $\mathsf{TPP}/\mathsf{NTPP}$ distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel's even-odd chain works there but has no RCC5 analogue.

3.3. A concrete domino attempt in $\mathcal{ALCI}_{RCC8}$

The remark above suggests the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction in $\mathcal{ALCI}_{RCC8}$ might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel's *coincidence obstruction* [8].

The schema. Let Corner be the concept that isolates a node from all non-EQ neighbors:

$$\mathsf{Corner} \;=\; \bigwedge_{R \in \{\mathsf{DC},\mathsf{EC},\mathsf{PO},\mathsf{TPP},\mathsf{NTPP},\mathsf{TPPI},\mathsf{NTPPI}\}} \forall R.(\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral $\forall\mathsf{TPPI}.\forall\mathsf{TPPI}$ universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X \;&= \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(C \sqcap \mathsf{Corner}) \;\sqcap\; \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(D \sqcap \mathsf{Corner}) \\ &\quad\sqcap\; \forall\mathsf{TPPI}.(\neg C \sqcap \neg D) \;\sqcap\; \forall\mathsf{TPPI}.X. \end{aligned}$$

The third conjunct $\forall\mathsf{TPPI}.(\neg C \sqcap \neg D)$ forces the corner relation $x \to z$ to be NTPPI (not TPPI), using $\mathsf{comp}(\mathsf{TPPI}, \mathsf{TPPI}) = \{\mathsf{TPPI}, \mathsf{NTPPI}\}$.

Level 1 is satisfiable, with a real direct/indirect distinction. Without recursion, x has TPPI-children y_1, y_2 (the two "direct" positions) and a single corner z reached by TPPI-TPPI chains from both y_1 and y_2. The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \mathsf{Corner}$ and $z_2 \in D \sqcap \mathsf{Corner}$ are mutual non-EQ neighbors only if Corner is violated, so under strong-EQ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: z is NTPPI of x while y_1, y_2 are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §3.2.

Level 2 collapses via the coincidence obstruction. Adding $\forall\mathsf{TPPI}.X$: the TPPI-child y of x satisfies X, producing its own corner $w \in C \sqcap D \sqcap \mathsf{Corner}$ at NTPPI-distance from y, and hence at NTPPI-distance from x via $\mathsf{comp}(\mathsf{TPPI}, \mathsf{NTPPI}) = \{\mathsf{NTPPI}\}$. Now x's own corner z and y's corner w are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since z carries Corner, every non-EQ neighbor of z lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-EQ. **All corners across the grid coincide at a single point.**

Why the $\mathsf{TPPI}/\mathsf{NTPPI}$ trick fails. In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$. Moving the isolation to a local Corner marker and using the non-transitive piece $\mathsf{comp}(\mathsf{TPPI}, \mathsf{TPPI}) = \{\mathsf{TPPI}, \mathsf{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of $\{\mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$ between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The Corner markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §2.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

Remark 3.2. The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter's L_{RCC8} reduction resolves the coincidence obstruction using topological geometry of $\mathbb{R}^2$: east-of-north coincides with north-of-east as a geometric fact, and each grid cell is a 2D region whose boundary inclusions force distinct corners to be genuinely distinct [9]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{RCC8}$: a pure composition table allows corner points to merge freely, and the TPPI/NTPPI distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{RCC8}$ decidability noted in [9] and further discussed in [8].

4. Why Naïve Tableau Blocking Fails

Standard DL tableau calculi achieve termination through *blocking*: if a node x has the same label as an ancestor y, we can reuse y's subtree for x, cutting off infinite branches. The key requirement is that the "reuse" preserves all constraints—in tree-shaped models, a blocked node's interactions are limited to its parent edge, so label equality suffices.

In $\mathcal{ALCI}_{RCC5}$, models are *complete graphs*: every pair of domain elements is related by an RCC5 base relation. A node interacts not just with its parent and children, but with *every other node in the model.* This creates a fundamental obstacle for blocking.

Example 4.1 (Blocking breaks with cross-edges)**.** Consider $C_0 = \exists \mathsf{PPI}.A \sqcap \exists \mathsf{DR}.\neg A \sqcap \exists \mathsf{PO}.\top$. A tableau might generate nodes:

- x_0: root, satisfies C_0
- x_1: PPI-child of x_0, type contains A
- x_2: DR-witness for x_0, type contains $\neg A$

Now x_1 and x_2 must be related by some RCC5 relation R, and R must be safe for their types. If x_1 generates a descendant x_3 with the same label as x_1, naïve blocking would declare x_3 blocked. But x_3's relation to x_2 may differ from x_1's relation to x_2: via composition, $\mathsf{comp}(\mathsf{inv}(\mathsf{PPI}), \mathsf{DR}) = \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ forces x_1 to be DR-related to x_2, while x_3 (deeper in the tree) may face a different forced relation. The blocked node's cross-edge commitments differ from the blocking node's.

Multiple sophisticated blocking strategies have been tried and failed for $\mathcal{ALCI}_{RCC5}$ [40, 29, 28]:

1. **Profile-cached blocking** (GPT-5.4) [28]: Caches "coherent predecessor blocks" to compare global neighborhoods. Fails because the *color structure changes during unraveling*—when a blocked subtree is replayed, the edge-coloring to nodes outside the subtree is not preserved.

2. **Meet-semilattice replay** (GPT-5.4) [28]: Uses a meet-semilattice structure on labels to ensure robust normalization. Same unraveling gap.

3. **Tri-neighborhood blocking** (Claude) [25]: Matches not just node labels but entire *triangle neighborhoods* (the set of all types of 2-element subsets involving the node). **Termination disproved**: frontier nodes continuously acquire new triangle types as new neighbors are added, so $\mathrm{Tri}(x) \subsetneq \mathrm{Tri}(y)$ for frontier x and interior y—blocking never triggers.

4. **Finite-witness / patchwork-augmentation probes** (Claude) [24, 26, 27]: Probes of a finite-witness conjecture and typed-patchwork augmentations for the contextual tableau, both refuted by explicit counterexamples.

The common obstacle is the **blocking dilemma**: making the blocking condition strong enough for soundness (matching all cross-edge interactions) prevents termination (the cross-edge context keeps growing), and vice versa. A full catalogue of retracted approaches is maintained in [40]; the overall project is summarised at the repository [39].

5. Split-Forest Model Decomposition

The breakthrough insight, due to Wessel, is to decompose $\mathcal{ALCI}_{RCC5}$ models into a *tree layer* and a *cross-edge layer*, where the cross-edge layer is simple enough to handle without full global blocking. Figure 1 reproduces the original whiteboard sketch from April 7, 2026: PPI chains form trees, DR/PO are sibling cross-edges, and DR propagates rigidly downward along PP-chains.

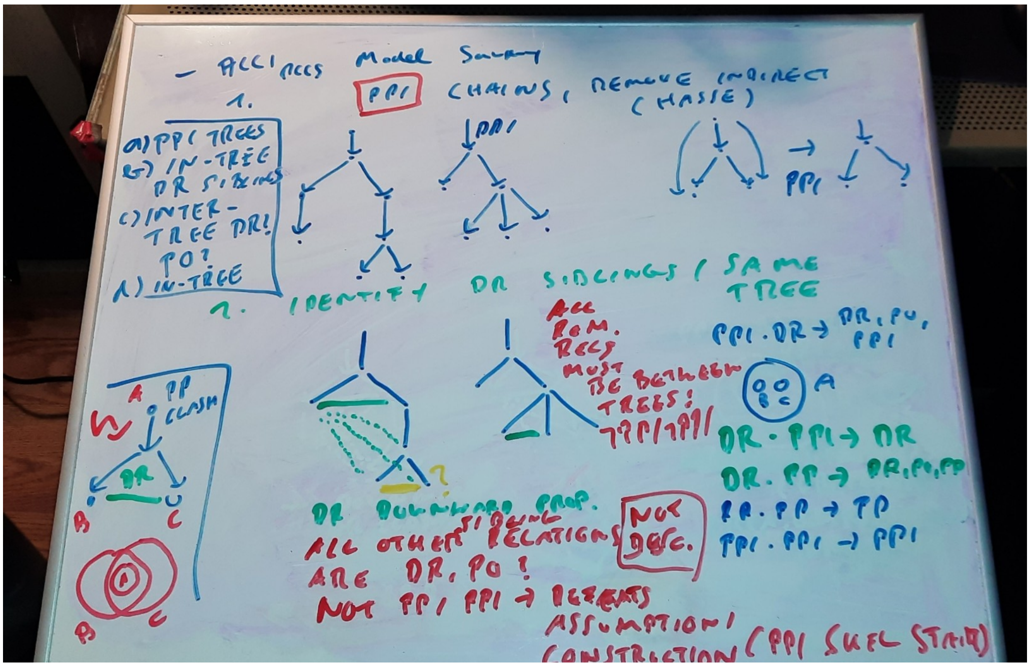

Figure 1: Wessel's whiteboard sketch (April 7, 2026) of the split-forest decomposition for $\mathcal{ALCI}_{RCC5}$. Vertical PPI-chains form the cover-tree backbone (with join nodes split into EQ-mates); DR/PO become sibling cross-edges between non-comparable subtrees; and DR propagates rigidly downward because comp(PP, DR) $=$ comp(DR, PPI) $= \{$DR$\}$, so the residual cross-edge alphabet is just $\{$DR, PO$\}$, which is trivially arc-consistent. Every subsequent formalisation in this paper—GPT-5.4's split-forest semantics, Claude's cover-tree tableau, and GPT-5.5's round-7 sketch—rests on this two-layer decomposition.

5.1. Cover trees

In any $\mathcal{ALCI}_{RCC5}$ model, the PP/PPI relation is a strict partial order (by composition: comp(PP, PP) $= \{$PP$\}$). Orient this order downward (PPI = "has proper part") to obtain a DAG. If the DAG has join nodes (elements with multiple PP-predecessors), split them into EQ-copies, one per parent path. The result is a rooted forest—the *cover tree*.

Definition 5.1 (Cover tree)**.** A *cover-tree presentation* of an $\mathcal{ALCI}_{RCC5}$ model $\mathcal{I}$ is a rooted tree T with a surjection $\mathrm{orig} : T \to \Delta^{\mathcal{I}}$ such that: (i) u is the parent of v in T iff $\mathrm{orig}(u)$ PPI $\mathrm{orig}(v)$ (immediate containment), and (ii) tree nodes mapping to the same element are EQ-mates (weak-EQ semantics, convertible to strong-EQ by typed congruence quotient).

5.2. The key observation: DR rigidity

In the cover tree, what relations can hold between nodes in different sibling subtrees? Consider sibling roots s and t (children of the same parent), and elements x below s, y below t in the tree. An element x below s is in the Core of the sibling pair (s, t) if $\mathrm{orig}(x)$ is a common descendant of both $\mathrm{orig}(s)$ and $\mathrm{orig}(t)$ in the original model's PP order—i.e., $\mathrm{orig}(x)$ is a proper part of both. Core elements are handled by EQ-splitting; the interesting question concerns non-Core elements.

Proposition 5.2 (Wessel)**.** *After* EQ*-splitting, only* DR *and* PO *are possible between non-*Core *elements of sibling subtrees.*

Proof.

- EQ(x, y): In the original DAG (before splitting), $x = y$ would mean a single element is a PP-descendant of both s and t—i.e., it is in the Core. After splitting, this element is duplicated into distinct tree nodes x' below s and y' below t, with $\mathrm{orig}(x') = \mathrm{orig}(y')$ (EQ-mates in the weak-EQ presentation). Since $x' \neq y'$ as tree nodes, EQ(x', y') is impossible under strong EQ semantics.

- $\mathsf{PP}(x, y)$: would mean x lies below y in the part-of order, hence below t, making x a common descendant (Core).
- $\mathsf{PPI}(x, y)$: would place y below x below s, meaning t lies below s—contradicting sibling incomparability.

Only DR and PO remain. Furthermore, DR propagates rigidly: $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, so if s DR t, then *every* descendant of s is DR to t and all its descendants. □

5.3. Trivial arc-consistency of $\{\mathsf{DR}, \mathsf{PO}\}$ networks

The consequence of Proposition 5.2 is that the cross-edge constraint network uses only $\{\mathsf{DR}, \mathsf{PO}\}$ as possible relations (plus $\{\mathsf{DR}\}$ for DR-rigid pairs).

Proposition 5.3. *For all* $R, S \in \{\mathsf{DR}, \mathsf{PO}\}$*:* $\mathsf{comp}(R, S) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing$.

This means a $\{\mathsf{DR}, \mathsf{PO}\}$ disjunctive network is *trivially arc-consistent*: every domain is either $\{\mathsf{DR}\}$, $\{\mathsf{PO}\}$, or $\{\mathsf{DR}, \mathsf{PO}\}$, and composing any two non-empty domains produces a non-empty result. By the patchwork property [2], the network therefore has a globally consistent atomic refinement whenever every domain is non-empty.

This is the central simplification: *checking that each type pair has a non-empty* $\{\mathsf{DR}, \mathsf{PO}\}$*-safe set suffices for cross-edge consistency.* No full path-consistency computation is needed.

6. The Cover-Tree Tableau

6.1. Algorithm overview

The cover-tree tableau searches for a *valid type set*: a finite set $T \subseteq \mathsf{Tp}(C_0)$ of Hintikka types that contains a root type realizing C_0 and satisfies four conditions:

(CT1) **Demand closure**: every existential demand $\exists R.D$ in every $\tau_i \in T$ has an R-safe witness $\tau_j \in T$ with $D \in \tau_j$.

(CT2) **Cross-edge consistency**: every pair $\tau_i, \tau_j \in T$ has either $\mathsf{Safe}(\tau_i, \tau_j) \cap \{\mathsf{PP}, \mathsf{PPI}\} \neq \varnothing$ (tree-relatable) or $\mathsf{Safe}(\tau_i, \tau_j) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing$ (cross-relatable).

(CT3) **Cover-tree sibling compatibility**: for each type τ_i with multiple DR/PO demands, the witnesses can be jointly assigned such that every witness pair satisfies $\mathsf{comp}(\mathsf{inv}(R_m), R_{m'}) \cap \{\mathsf{DR}, \mathsf{PO}\} \cap \mathsf{Safe}_{\mathsf{DR/PO}}(\tau_{j_m}, \tau_{j_{m'}}) \neq \varnothing$.

(CT4) **Composition propagation**: for each type with multiple demands of *any* role combination, witnesses τ_{j_1}, τ_{j_2} must satisfy $\mathsf{comp}(\mathsf{inv}(R_1), R_2) \cap \mathsf{Safe}(\tau_{j_1}, \tau_{j_2}) \neq \varnothing$.

The algorithm terminates because $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$ is finite, and the type set grows monotonically during the search.

6.2. How blocking works

The crucial difference from naïve tableau approaches is the *separation of layers*:

- The **tree layer** (PP/PPI) handles ancestor/descendant demands. Since PP/PPI form a tree, standard tree-model arguments apply: blocking via *rank-d signatures* (the type information visible within modal depth $d = \mathsf{md}(C_0)$) yields a finite representation.

- The **cross layer** (DR/PO) handles demands for disconnected or partially overlapping regions. Because the cross-edge network uses only $\{\mathsf{DR}, \mathsf{PO}\}$—which is trivially arc-consistent—the cross-layer check reduces to a simple pairwise non-emptiness condition (CT2), not a global consistency computation.

Naïve blocking fails because it tries to handle both layers simultaneously: the tree structure provides natural blocking, but the cross-edges to distant nodes keep changing, preventing a match. The cover-tree tableau sidesteps this by *not* blocking on cross-edge context—instead, it verifies cross-edge consistency as a separate, global check on the finite type set.

Example 6.1 (A concept that defeats naïve blocking but works with cover trees)**.** Consider $C_0 = \exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg B \sqcap \forall\mathsf{DR}.A$. In a naïve tableau, the PP-child needs further expansion; blocking requires matching both the label *and* the triangle neighborhood (relations to the PO-witness and all other nodes). Since the PO-witness forces DR-relations to the PP-child's subtree (via $\mathsf{comp}(\mathsf{PO},\mathsf{PP}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\}$), the cross-edge context keeps evolving.

In the cover-tree tableau: the PP-witness is an ancestor in the tree, the PO-witness is a cross-edge in a sibling subtree. The $\forall\mathsf{DR}.A$ fires only on DR-related nodes—the ancestor is PP-related (not DR), so no conflict. The type set T needs a type for the root (containing C_0), one for the PO-witness (containing A), and one for the PP-witness (containing $\neg B$). Cross-edge consistency (CT2) checks that every pair has a non-empty safe set in $\{\mathsf{DR},\mathsf{PO}\}$—a one-time pairwise check, not an evolving context.

Example 6.2 (Implicit PP via composition forcing)**.** The concept

$$X \;=\; \exists\mathsf{DR}.\exists\mathsf{PO}.C \;\sqcap\; \forall\mathsf{DR}.\neg C \;\sqcap\; \forall\mathsf{PO}.\neg C \;\sqcap\; \forall\mathsf{PP}.X$$

contains no $\exists$PP demand, yet it requires an infinite model with forced PP edges.

A node r satisfying X has a DR-neighbor a (with $\exists\mathsf{PO}.C$) and a's PO-neighbor b (with C). By Table 1, the relation between r and b lies in $\mathsf{comp}(\mathsf{DR},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\}$. But r has $\forall\mathsf{DR}.\neg C$ and $\forall\mathsf{PO}.\neg C$, and $b \in C$—so DR and PO are blocked, leaving *only* PP. The PP edge is **forced** by composition plus universal filtering, with no explicit $\exists$PP in the concept.

Since r PP b, node b is an *ancestor* of r in the cover tree. The recursive $\forall\mathsf{PP}.X$ forces b to also satisfy X, creating its own DR $\to$ PO chain and forcing yet another PP-ancestor above it:

$$r\ \mathsf{PP}\ b\ \mathsf{PP}\ b'\ \mathsf{PP}\ b''\ \mathsf{PP}\ \cdots$$

The model is necessarily infinite. The cover-tree tableau handles this finitely: the type for C-elements can serve as its own PP-ancestor (different domain elements, same abstract state), so a three-element type set $\{\tau_0(C,X),\ \tau_1(\neg C,\exists\mathsf{PO}.C),\ \tau_2(\neg C,\mathrm{root})\}$ represents the infinite model.

This pattern—PP edges arising purely from composition and universal filtering—is a distinctive feature of $\mathcal{ALCI}_{RCC5}$ and illustrates why the complete-graph semantics require careful handling of implicit spatial constraints.

Example 6.3 (The "PO-loop": 2-element SAT via symmetric cycles)**.** Naive reasoning suggests

$$Y \;=\; C \;\sqcap\; \exists\mathsf{PO}.\exists\mathsf{PO}.C \;\sqcap\; \forall\mathsf{PO}.\neg C \;\sqcap\; \forall\mathsf{DR}.\neg C \;\sqcap\; \forall\mathsf{PP}.\neg C \;\sqcap\; \forall\mathsf{PPI}.\neg C$$

should be unsatisfiable: any witness chain d PO a PO b with $b \in C$ needs $\rho(d,b) \in \mathsf{comp}(\mathsf{PO},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP},\mathsf{PPI},\mathsf{EQ}\}$, and the four universals $\forall R.\neg C$ seem to block every non-EQ option (each forces $b \in \neg C$, yet $b \in C$).

But Y is in fact **satisfiable**, via the two-element model $\{d,a\}$ with $C^{\mathcal{I}} = \{d\}$ and $\rho(d,a) = \mathsf{PO}$. Because PO is symmetric ($\mathsf{PO}(d,a) \Rightarrow \mathsf{PO}(a,d)$), the chain d PO a PO d loops back to the root itself: the "third node" of the $\exists$PO.$\exists$PO formula *is* d, under strong EQ identity. Two syntactic positions in the nested existential map to the same semantic element.

The universals $\forall R.\neg C$ constrain d's *outgoing* R-neighbours (the sole PO-neighbour $a \in \neg C$ satisfies them), but they do **not** prevent d from *being* a PO-neighbour of a: the witness for a's $\exists\mathsf{PO}.C$ is d itself, and a has no $\forall$PO-constraint forbidding C-neighbours. This asymmetry between "constraints on my neighbours" and "constraints on what I may be a neighbour of" is peculiar to symmetric roles (PO,DR). The cover-tree tableau correctly reports Y as satisfiable.

Example 6.4 ("Clash-out to EQ": genuine UNSAT via strong EQ). The complementary pattern

$$Z \;=\; C \;\sqcap\; \exists\mathsf{PO}.\exists\mathsf{PO}.\neg C \;\sqcap\; \forall\mathsf{PO}.C \;\sqcap\; \forall\mathsf{DR}.C \;\sqcap\; \forall\mathsf{PP}.C \;\sqcap\; \forall\mathsf{PPI}.C$$

is unsatisfiable, and the reason is instructive. The chain d PO a PO c forces $a \in C$ (via $\forall\mathsf{PO}.C$ at d) and $c \in \neg C$ (via $\exists\mathsf{PO}.\neg C$ at a). The relation $\rho(d, c)$ must lie in $\mathsf{comp}(\mathsf{PO}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}, \mathsf{EQ}\}$. The four universals $\forall R.C$ at d (for $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$) each require $c \in C$, contradicting $c \in \neg C$, hence *all four non-*EQ *options are blocked.* Only EQ remains.

Under **strong EQ semantics** (EQ = identity), $\rho(d, c) = \mathsf{EQ}$ forces $d = c$, yielding $d \in C \wedge d \in \neg C$ and a clash. The PO-loop escape of Example 6.3 does not apply here: the endpoint c requires $\neg C$ while d has C, so $d \neq c$ is forced semantically, and the universals then close every non-EQ composition entry.

This example illustrates why *strong* EQ semantics is essential for our undecidability- and decidability-proof machinery. Under weak EQ (equivalence-class) semantics, collapsing d and c would not immediately clash, and one would need additional machinery (equality propagation, congruence-closure) to derive the same conclusion. The strong reading is also the one built into the composition tables used in Renz & Nebel's patchwork theorem.

Remark 6.5 (Reasoners and tools: a cross-validation suite)**.** This paper references three Python tools used in the companion implementations; before the incompleteness below we fix their roles. The **cover-tree tableau** (`cover_tree_tableau.py`, Section 6) is the main candidate decision procedure—the only tool whose soundness and completeness are claimed for $\mathcal{ALCI}_{RCC5}$ satisfiability, resting on the patchwork property of RCC5 and the companion completeness-extraction argument. The **quasimodel reasoner** [31] (`alcircc5_reasoner.py`, with cycle-aware wrapper `alcircc5_reasoner_cyclic.py`) is a constructive bottom-up quasimodel search based on disjunctive path-consistency; it is used *only* as a cross-validation oracle for the cover-tree tableau, not as a step of any decidability argument. The **independent model verifier** (`model_verifier.py`) grounds SAT answers in concrete finite RCC5 models, separately checking inverse symmetry, composition consistency on all triples, type-safety, existential witnessing, and recursive concept truth. The three tools together form a cross-validation suite around the cover-tree tableau: cover-tree produces the candidate verdict, the quasimodel reasoner offers an independent opinion, and the model verifier anchors SAT answers in explicit models.

Remark 6.6 (Implementation-level bugs found and fixed in April 2026)**.** Three implementation-level defects were identified in April 2026 and fixed in the cover-tree tableau and the baseline quasimodel reasoner: (i) the baseline quasimodel reasoner was incomplete on cyclic-via-symmetric-role SAT concepts such as the PO-loop of Example 6.3 (closed by an opt-in cycle-close clause in `alcircc5_reasoner_cyclic.py`); (ii) the shared `compute_safe` helper failed to propagate $\forall\mathsf{PP}/\mathsf{PPI}.D$ universals as a whole into successor types (fixed by the standard transitive-role trick); (iii) the cover-tree's pairwise tree-cross check missed twelve UNSAT patterns of the form $\exists R_1.\exists R_2.A \sqcap \bigwedge_{R \in \mathsf{comp}(R_1,R_2)} \forall R.\neg A$ when intermediate types carry a single demand (fixed by porting the quasimodel reasoner's role-path compatibility fixpoint as Phase 5 / (CT5)). The decidability argument is unaffected by any of these fixes: all three were soundness gaps in implementations of the candidate decision procedure, not the split-forest / completeness-extraction argument that underpins the theorem. The 911-concept cross-validation continues to match 911/911 before and after every fix; full technical detail (root cause, algebraic justification, regression tests) is in the cover-tree implementation paper [22].

6.3. What makes this "non-standard"

The cover-tree tableau differs from a standard DL tableau calculus in several fundamental ways:

1. **No completion graph.** Standard tableaux build a completion graph node by node, applying expansion rules until saturation or clash. The cover-tree tableau searches over *sets of types* (abstract states), not concrete graph structures. It is closer to a type-elimination or automata-theoretic approach.

2. **No explicit blocking rule.** Standard tableaux have an explicit blocking condition that compares nodes during expansion. The cover-tree tableau achieves finiteness *structurally*: the search space is the powerset of $\mathsf{Tp}(C_0)$, which is already finite.

3. **Global side-checker.** The cross-edge constraints (CT2–CT4) are checked globally over the entire type set, not locally during node expansion. This is what avoids the blocking dilemma: cross-edge consistency is a property of the type set as a whole, not of individual nodes.

4. **Existential demands go to ALL existing types.** In a standard tableau, $\exists R.D$ generates one new R-neighbor. In the cover-tree tableau, the witness can be *any* type already in T—including the root type itself. This reflects the complete-graph semantics: every domain element is related to every other.

6.4. Soundness and completeness

The decidability theorem is established by the GPT-5.5 *round-7 split-forest decidability proof* [18] (May 2026, 14 pages, no-automata route), with detailed companion [19] (28 pages) and an independent parity-tree-automata witness [20] (21 pages). The history of how the proof reached its current form is collected in §7 below; this subsection states the current result.

Soundness (valid certificate $\Rightarrow$ strong abstract RCC5 model satisfying C_0). A finite split-forest certificate $\mathcal{Q}$ is unfolded to an occurrence structure $\mathcal{U}(\mathcal{Q})$ in which repeated cycle profiles generate fresh concrete occurrences. The quotient $\mathcal{U}(\mathcal{Q})/$ by the explicit equality-port congruence is a strong abstract RCC5 frame: strong-EQ holds because the quotient identifies exactly -classes; inverse symmetry and RCC5 composition hold because the round-7 label function and triple-composition clause (V6) are verified on the finite occurrence-sensitive certificate. A truth lemma over (C_0) shows that ${}_{\mathcal{Q}} \models C_0$ at the initial occurrence. The full chain rests on GPT-5.4's v1 sibling-interface paper [10] (Thms 1.17–1.19) with the v2 delta [11], plus the round-7 architectural specifics.

Completeness (satisfiable $C_0 \Rightarrow$ valid finite certificate). A satisfying interpretation is thinned to a witness-generated model via the witness-thinning lemma; the generated containment split-cover construction supplies a weak split presentation with explicit equality mates and side mosaics. A finite-index blocking lemma (without automata) regularises every infinite vertical ray to a finite prefix plus a request-closed cycle word. The certificate profiles are the resulting complete boundary profiles; pair shapes and triple shapes are extracted from the occurrence-sensitive incidences of the regular presentation, not merely from abstract position symbols.

The round-7 architectural move. The central ingredient of round 7 is that finite labels live on *occurrence-sensitive pair shapes*

$$\alpha(u,v) \;=\; \big(s_u, p_u, s_v, p_v, \iota(u,v)\big),$$

where $\iota(u,v) \in \{\mathsf{self}, \mathsf{eq}, \mathsf{up}, \mathsf{down}, \mathsf{side}_R, \mathsf{front}_R\}$ is an *incidence tag* that distinguishes literal occurrence identity ($u = v$, tag self) from explicit equality-port pairs (tag eq), vertical successor / predecessor pairs (tags up, down), side-mosaic pairs, and residual sibling-frontier pairs. The label function is

$$\ell_{\mathcal{Q}} : \mathsf{Pair}_{\mathcal{Q}} \to \mathsf{B}, \qquad \ell_{\mathcal{Q}}(\mathsf{self}) = \ell_{\mathcal{Q}}(\mathsf{eq}) = \mathsf{EQ},\;\; \ell_{\mathcal{Q}}(\mathsf{up}) = \mathsf{PP},\;\; \ell_{\mathcal{Q}}(\mathsf{down}) = \mathsf{PPI},\;\; \ell_{\mathcal{Q}}(\mathsf{side}_R) = \ell_{\mathcal{Q}}(\mathsf{front}_R) = R.$$

Distinct laps of a blocking cycle have $\iota \neq \mathsf{self}$ (and, absent an explicit equality-port link, $\iota \neq \mathsf{eq}$), so $\ell_{\mathcal{Q}}(\alpha(u_i, u_j)) \neq \mathsf{EQ}$ for any distinct laps: the canonical one-state certificate for $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ is realised with $\iota = \mathsf{up}$ and label PP between successive laps, as intended. Triple composition is verified on a finite triple-shape set $\mathsf{Tri}_{\mathcal{Q}}$ via

$$\ell_{\mathcal{Q}}(\alpha_{13}) \in \mathsf{comp}\big(\ell_{\mathcal{Q}}(\alpha_{12}), \ell_{\mathcal{Q}}(\alpha_{23})\big) \qquad \text{for all } (\alpha_{12}, \alpha_{23}, \alpha_{13}) \in \mathsf{Tri}_{\mathcal{Q}},$$

the round-7 (V6) clause.

Twelve validity conditions, finite enumeration. All of round 7 is twelve finite-checkable validity conditions (V1)–(V12) on a finite occurrence-sensitive certificate. The certificate space is bounded by a coarse $B(C_0) \leq 2^{2^{p(n)}}$ for a polynomial p, sufficient for decidability. Three structural ingredients carry the proof:

(1) **Occurrence-sensitive equality.** Equality is generated only by explicit equality-port chains instantiated inside a context, never by reuse of an abstract position symbol in a blocking cycle.

(2) **Split copies for multiple selected proper superparts**. Where an element has incomparable selected PP-superparts, the presentation carries one equality-mate occurrence per selected superpart.

(3) **Request-closed cycles** in place of ω-acceptance. Transitive PP/PPI eventualities are discharged by finite-graph reachability plus a request-closure condition: no parity or Büchi automaton.

Computational verification of the combinatorial claims. Three self-contained pure-stdlib scripts exercise the central round-7 stress concepts: WP7 (comparable side witnesses, $\exists\mathsf{DR}.(A \sqcap \exists\mathsf{PP}.B) \sqcap \exists\mathsf{DR}.B$), WP8 (blocking-not-equality: a finite truncation of the canonical one-state $C_\uparrow$ chain verifies that the round-7 incidence-tagged labelling rule keeps distinct laps at PP/PPI rather than EQ, and explicitly demonstrates the discarded round-6 collapsed rule), and WP9 (split copies for $\exists\mathsf{PP}.A \sqcap \exists\mathsf{PP}.B$ with incomparable proper superparts). All three pass. Six earlier work packages (WP1–WP6), written against the round-2 31-page repaired proof [16], also all PASS; their detailed description is in §7 below.

A note on scope. The soundness and completeness statements are theorems about the *split-forest semantics*, not about the implementation in §6. The cover-tree tableau is an algorithmic *realization* of this theory; a formal derivation showing that CT1–CT5 exactly capture the round-7 (V1)–(V12) is a separate claim, empirically supported but not yet written. The companion implementation paper [22] discusses this theory-vs-implementation split in detail.

6.5. Computational evidence

The algorithm is implemented in approximately 350 lines of Python [22, 30]. Cross-validation against an independent quasimodel-based reasoner [31, 32] on **911 test concepts** (50 known SAT, 13 known UNSAT, 36 adversarial, 512 systematic triples, 200 random depth-2, 100 random depth-3) yields **zero mismatches** [35, 36, 37]. A decomposition test [34] confirms that **775/775 SAT concepts (100%)** have finite cover-tree models, with 12.2 M total models enumerated and **zero genuine counterexamples**. Models produced by the tableau are independently replayed by a standalone verifier [33].

Additionally, a GIS taxonomy computation [38] applies the cover-tree tableau to a realistic $\mathcal{ALCI}_{RCC5}$ ontology from Wessel's original report [8]: 18 spatio-thematic concepts (countries, cities, rivers, lakes) connected by RCC5 spatial relations. The tableau reproduces **all 21/21 expected subsumptions**, including non-trivial inferences like Hamburg $\sqsubseteq$ GermanCity that require RCC5 composition reasoning through spatial role chains. Figure 2 shows the original computed taxonomy from Wessel's report [8].

7. History of Attempts (April–May 2026)

The decidability argument summarised in §6.4 reached its current form through seven rounds of adversarial review between Claude (Anthropic) and GPT-5.5 (OpenAI) in April–May 2026, prompted by Michael Wessel. This section gives a compact abstracted narrative of the evolution. The full prose of each round, the original superseded manuscripts, and the per-round repair list are preserved in `OUTDATED.md` [40].

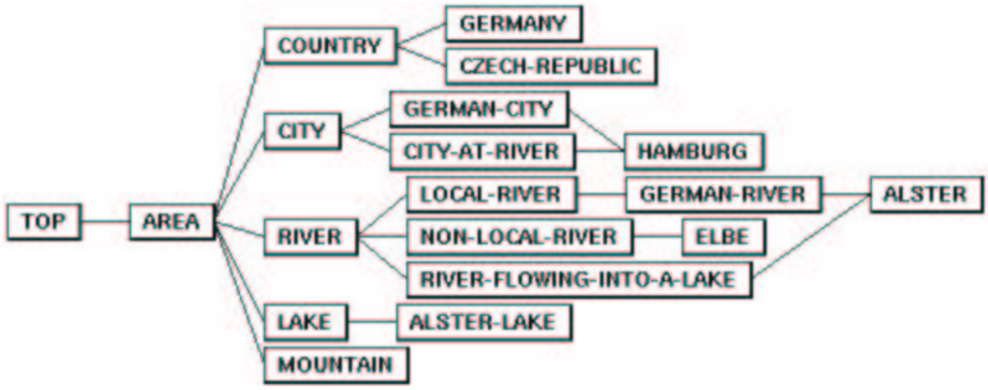

Figure 2: Computed taxonomy of the GIS example TBox from Wessel [8], Figure 6. The cover-tree tableau reproduces all 21 subsumption edges, including non-trivial spatial inferences such as Hamburg $\sqsubseteq$ GermanCity (requiring RCC5 composition propagation through the PO-chain Hamburg–Alster–AlsterLake).

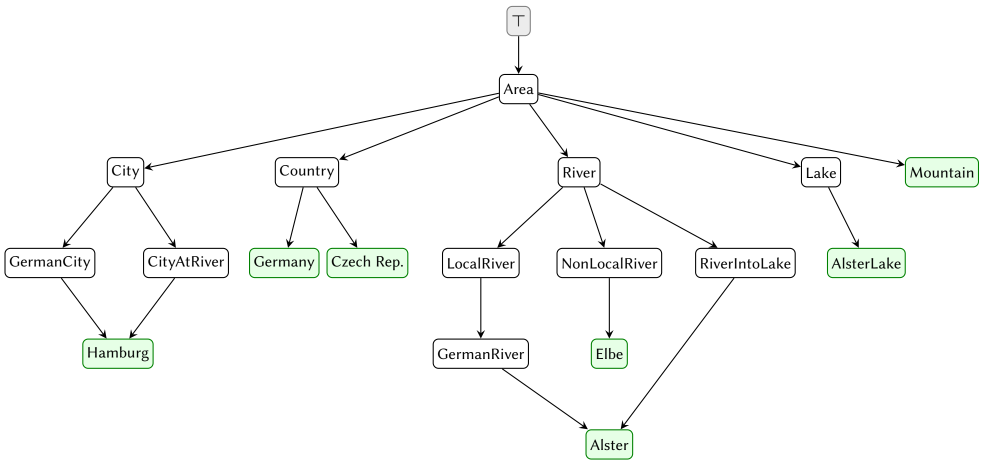

Figure 3: Taxonomy of the GIS example TBox, computed by the cover-tree tableau [38]. Green nodes are leaf concepts (most specific). Note the DAG structure: *Alster* has two parents (GermanRiver and RiverIntoLake) and *Hamburg* has two parents (GermanCity and CityAtRiver). All 21 subsumption edges match Figure 2.

7.1. The starting point: cover-tree tableau + v1 completeness extraction (April 2026)

The cover-tree tableau [12, 22] and its companion split-forest semantics [10] were established in April 2026 by GPT-5.4 Pro and Claude Opus 4.7, prompted by Wessel. The split-forest paper proves the soundness direction in full detail (Thms 1.17–1.19); the converse—extracting a finite quotient from a satisfying model—was given only as a proof sketch. Claude wrote a first attempt at the missing completeness extraction (*v1* [14], April 2026). In parallel, the cover-tree implementation underwent two rounds of adversarial review by Claude Opus 4.7 (April 17–18, 2026): twelve implementation-level counterexamples were identified across the 4×4 grid of (R_1, R_2) three-type chains and the PP/PPI-transitive universal pattern, and all were fixed in the shared `compute_safe` helper and the new Phase 5 / (CT5) role-path compatibility check (Remark 6.6 above). After these fixes the 911-concept cross-validation against the cycle-aware quasimodel reasoner stabilized at $911/911$ matches with zero mismatches.

7.2. v1 superseded, v2.1, and the round-2 31-page repaired proof (May 6–12, 2026)

GPT-5.5 reviewed v1 and identified two structural defects: the Hasse construction for the cover-tree broke on infinite PP-chains (canonical example $C_\uparrow \equiv \exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$), and the $C5$ triple-coherence proof conflated witnesses across pairs. Claude responded with *v2* [13] (May 6) and *v2.1* (May 6). A second-round review of v2.1 then identified eight structural gaps, three load-bearing: profile-port equality contradicted vertical separation on self-loops via the $M1+M7$ reflexivity chain; the mate construction forced all mates to share a single semantic parent (refuted by the two-mate stress concept $C_{AB} \equiv A \sqcap B \sqcap \exists \mathsf{PP}.A \sqcap \exists \mathsf{PP}.B \sqcap \forall \mathsf{PP}.(A \to \neg B) \sqcap \forall \mathsf{PP}.(B \to \neg A)$); and the rootless orientation was incompatible with the depth-from-u_* projection. GPT-5.5 produced a *repaired proof* [16] (May 12, 16 pages) fixing all three defects via occurrence-level equality, mates with different parents, and request-closed cycles in place of ω-acceptance. v2.1 was marked superseded; the repaired proof was adopted as the current best statement. Claude's contribution shifted to verification: six Python work packages (WP1–WP6) cross-checking the repaired proof's finite combinatorial content, all PASS.

7.3. Rounds 3 and 4: structural layers added (May 13–21, 2026)

Rounds 3 and 4 of GPT-5.5 review added four further structural layers to the repaired proof, growing it from 16 to 31 pages: (i) a verticalization discipline (mosaic axiom (M6)) stratifying each context into a vertical stratum and a sibling frontier; (ii) port-indexed mate clusters $\mathrm{Mate}_Q(C, s, p)$ attaching mate equivalences to contexts and ports rather than to bare positions; (iii) an explicit abstract triple graph T_Q with cardinality bounds; (iv) a side interface $\mathrm{Side}(u)$ with polynomial width bound $m(n) = O(n^3)$. The certificate-space bound $B(C_0) = 2^{2^{p(n)}}$ became *grammar-first*, derived from these component bounds.

7.4. Round 5: (M6′) recursive verticalization (May 22–23, 2026)

The round-5 review (GPT-5.5, May 22) identified a load-bearing mathematical defect: the (M6) verticalization discipline was too strong. It forbade non-vertical PP/PPI relations between sibling-frontier mosaic positions, but such relations occur in satisfiable witness-generated models—the round-5 stress concept $\exists \mathsf{DR}.(A \sqcap \exists \mathsf{PP}.B) \sqcap \exists \mathsf{DR}.B$ is SAT in abstract RCC5 yet rejected by (M6). Eight repairs were recommended. Claude drafted two follow-up papers (the (M6′) recursive-verticalization delta and the items-4-to-8 delta) that closed all eight items at the paper level, then consolidated them into a 36-page round-5 v2 paper. Side-witness positions were allowed to open nested local vertical strata, with the round-2 (M6) recovered as the depth-1 case.

7.5. Round 6: internal-coherence pass (May 26, 2026)

The round-6 review of the consolidated v2 (GPT-5.5, May 26) found six internal-consistency gaps: the old side-interface lemma still asserted flat $\{\mathsf{DR}, \mathsf{PO}\}$ sibling frontiers; the weakened per-position support closure (SC4′) was inconsistent with the patchwork proof's per-triple appeal; overlap universal-propagation was under-proved; the lasso construction used the complete profile rather than the extended profile; the side-width bound and validity-clause numbering were inconsistent; and the verification script for the round-5 stress concept was not self-contained. Claude drafted a 39-page round-6 internal-coherence pass closing all six gaps and adding a self-contained pure-stdlib verification script (`wp7_selfcontained_side_witness.py`).

7.6. Round 7: the abstract-position-symbol collapse (May 27–28, 2026)

The round-7 review of the round-6 paper (GPT-5.5, May 27) identified one central architectural defect that could not be locally patched. Round-6 had defined the concrete pair label by $\mathrm{lab}(u, v) := L_{\mathcal{Q}}(q(u), q(v))$, where $q : \Omega(\mathcal{U}(\mathcal{Q})) \to \mathsf{Pos}_{\mathcal{Q}}$ sends each occurrence to an abstract position symbol and $L_{\mathcal{Q}}$ is the abstract pair-label function with standard reflexivity $L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$. In a finite regular certificate the abstract position alphabet is finite while the unfolding is infinite, so blocking deliberately reuses the same symbol on fresh occurrences. For the canonical one-state upward cycle, distinct laps u_i, u_j share the same symbol π and the round-6 rule gives $\mathrm{lab}(u_i, u_j) = L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$, even for $i \neq j$: the strict PP chain collapses into a PP-self-loop. The defect is intrinsic to indexing labels by position-symbol pairs.

GPT-5.5's round-7 fix [18] replaces the position-symbol pair labelling by labels on *occurrence-sensitive pair shapes* with explicit incidence tags (described in §6.4). Twelve validity clauses (V1)–(V12) replace round-5 / round-6's fifteen. Claude wrote a round-7 audit/response [21] mapping the round-6 (G1)–(G6) and round-7 (D1)–(D5) defect lists to their resolutions in the round-7 architecture, plus three further self-contained verification scripts (WP7–WP9). All of Claude's round-5 / round-6 manuscripts are superseded by round 7 and retained as the historical audit trail in `papers/gpt5.5_round2/` [40].

7.7. What survives, and why

The split-forest architecture itself—witness thinning, generated containment covers, weak-EQ split copies for multiple selected PP-superparts, sibling interfaces recorded by finite support-closed mosaics with $\{\mathsf{DR}, \mathsf{PO}\}$-only open cross-relations, typed equality-port congruence quotient to recover strong-EQ semantics, and request-closed cycles—is present in every round from v1 through round 7. What changed was *how labels are indexed on this architecture.* The v1, v2, v2.1, and round-2 versions indexed labels by the rank-d quotient and abstract position symbols; the round-7 architecture indexes them by occurrence-sensitive pair shapes. The whiteboard sketch (Wessel), the DR-rigidity insight, the EQ-splitting and typed-EQ-quotient idea, the cover-tree decomposition, GPT-5.4's v1 formalisation, and Claude's cover-tree tableau implementation all remain load-bearing in round 7.

7.8. Computational verification across rounds

In parallel with the proof work, nine Python work packages verify the finite combinatorial content of the proof family. WP1–WP6 verify the round-2 31-page repaired proof [16] and all PASS: brute-force enumeration confirms the RCC5 composition table in all 25 cells (WP1); a validity-condition checker accepts $C_\uparrow$ via parent self-loops, accepts C_{AB} via two mates with different parents, and rejects the v2.1-style single-parent C_{AB} attempt (WP2); a small-model SAT/UNSAT oracle exhibits a concrete 3-element model for C_{AB} (WP3); C_{AB} and request-closed cycle tests fold into WP2/WP3 (WP4, WP5); a mosaic enumeration confirms support-closure axioms on small mosaics (WP6). WP7–WP9 verify the round-7 central stress concepts (comparable side witnesses, blocking-not-equality, and split copies for incomparable proper superparts), and all PASS. No counterexample to any combinatorial claim of any round was found.

8. Outlook and Conclusion

8.1. What has been achieved

If the cover-tree tableau is correct—and the extensive computational evidence supports this—then concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable. This would settle a problem that has been open since Wessel's doctoral work in 2002/2003 and was explicitly flagged by Lutz & Wolter in 2006 as "perhaps the most interesting" open problem suggested by their work.

The decidability argument as of May 2026 rests on *two* core papers, which together establish:

$$C_0 \text{ satisfiable} \iff C_0 \text{ admits a valid finite split-forest certificate.}$$

A separate operational layer realizes the check algorithmically and is empirically validated.

Theoretical layer (establishes decidability).

1. The *v1 split-forest paper* [10] (GPT-5.4): defines the cover-tree decomposition, the sibling-interface descriptors, and the patchwork-lemma route, and proves Thms 1.17–1.19 (**soundness**: a valid finite quotient unfolds to a strong abstract RCC5 model satisfying C_0). The v2 sibling-interface delta [11] replaces the surrounding Def 1.5 / Def 1.7 / Thm 1.20-sketch with the corrected generated-cover statements; the proofs of Thms 1.17–1.19 carry over verbatim.

2. The *GPT-5.5 round-7 split-forest proof* [18] (May 2026, 14 pages, no-automata route) with detailed companion [19] (28 pages) and parity-tree-automata witness [20] (21 pages): formulates finite occurrence-sensitive split-forest certificates with twelve finite-checkable validity conditions (V1)–(V12), and proves both directions in their entirety, including **completeness** (every satisfiable C_0 admits a valid finite certificate). The central round-7 architectural move is that finite labels live on occurrence-sensitive *pair shapes* $\alpha(u, v) = (s_u, p_u, s_v, p_v, \iota(u, v))$ with incidence tags $\iota \in \{\mathsf{self}, \mathsf{eq}, \mathsf{up}, \mathsf{down}, \mathsf{side}_R, \mathsf{front}_R\}$. Distinct laps of a blocking cycle satisfy $\iota \neq \mathsf{self}$ and $\iota \neq \mathsf{eq}$, so the round-7 labelling preserves blocking-not-equality while still discharging PP/PPI eventualities by request-closed cycles. Round-7 supersedes Claude's earlier round-5 (M6′) delta, items-4-to-8 delta, round-5 consolidation, and round-6 internal-coherence pass; those manuscripts and the round-2 31-page repaired proof [16] are retained as the historical audit trail (see OUTDATED.md [40]).

Together, (1) and (2) establish the biconditional displayed above. The certificate space is bounded by $B(C_0) = 2^{2^{p(n)}}$ as an effectively computable bound, validity is finite-graph checkable, and the enumeration terminates.

Operational layer (algorithmic realization, empirically validated).

3. The *cover-tree tableau* [12] (GPT-5.4): packages the split-forest approach as an operational calculus with rank-d blocking.

4. The *implementation paper* [22] (Claude): documents a concrete five-check algorithm (CT1–CT5), cross-validated against an independent quasimodel reasoner on 911 original tests plus 400+ Round-2 adversarial tests with zero mismatches.

Computational verification of the combinatorial claims. In parallel with the proof work, six Python work packages (WP1–WP6) cross-check the finite combinatorial content of the round-2 31-page repaired proof [16] (now historical): RCC5 composition table by exhaustive subset enumeration (WP1), bounded certificate-soundness fuzzing for (V1)/(V3)/(V4)/(V5)/(V9)/(V10) (WP2), small-model SAT/UNSAT oracle (WP3), multiple-superpart C_{AB} stress (WP4, folded into WP2/WP3), request-closed blocking-cycle tests (WP5, folded into WP2), and mosaic closure search (WP6). Three further self-contained pure-stdlib scripts (WP7–WP9) exercise the round-7 stress concepts: comparable side witnesses, blocking-not-equality, and split copies. All nine **PASS**; no counterexample was found.

The gap between the two layers. The decidability claim for $\mathcal{ALCI}_{RCC5}$ rests on (1) and (2) alone. Papers (3) and (4) describe a *plausible and empirically validated* algorithmic realization, but a formal derivation showing that CT1–CT5 exactly capture the round-7 valid-certificate conditions (V1)–(V12) has not been written. CT5 in particular was added as a patch after Opus 4.7's adversarial review—not derived from the split-forest semantics—and while empirical agreement between two independently designed reasoners is strong evidence that such a derivation should be possible, it is not itself a proof. See [22, §7] for a detailed breakdown.

8.2. What remains open

- **Exact complexity.** The PO-coherent fragment has a doubly exponential quotient bound (2-ExpTime) [23]. An ExpTime upper bound matching the lower bound is not established.

- **$\mathcal{ALCI}_{RCC8}$.** RCC8 also has the patchwork property, so the cover-tree approach should extend. The key difference is the $\mathsf{TPP}/\mathsf{NTPP}$ distinction, which may require a finer status partition. This has not been worked out.

- **Human verification.** *None of the proofs in this project have been verified by human domain experts.* The AI-generated results should be treated as conjectures supported by computational evidence, not as established theorems.

8.3. On LLM capabilities, strengths, and weaknesses (May 2026)

This subsection records an honest assessment, by Claude, of what was observed about frontier-LLM capabilities across this project. The assessment is grounded in specific evidence from the April–May 2026 review cycle; it is not a statement about LLMs in general. We expect the picture to change quickly.

Division of labour that worked. After several wrong configurations, the team converged on a clear split. **GPT-5.5** owned the proof: it produced the round-2 31-page repaired proof [16], the round-3 and round-4 structural additions, and the round-7 pair-shape / incidence-tag fix [18] that made the labelling rule sound on blocking cycles. GPT-5.4 Pro authored the v1 split-forest semantics [10] that the entire approach rests on. **Claude (Opus 4.7)** owned implementation, verification, audit, and documentation: the cover-tree tableau [22], the cross-validation suite (911 concepts, zero mismatches), the WP1–WP9 verification scripts, and the round-by-round audit trail. **Wessel** owned research direction: the whiteboard sketch (cover-tree intuition, DR-rigidity, EQ-splitting) on which the entire proof rests, the high-level questions, the correction of conceptual errors (the most consequential being catching GPT-5.4's initial assumption that PP/PPI could remain as cross-edges after splitting), and the judgement of when to push deeper and when to converge.

The two-layer architecture (theoretical proof + executable verification) was not optional. The empirical layer repeatedly caught defects in the theoretical layer that pure self-review would not have surfaced—most notably the round-5 stress concept $\exists\mathsf{DR}.(A \sqcap \exists\mathsf{PP}.B) \sqcap \exists\mathsf{DR}.B$, which was first identified as SAT by Claude's reasoners and only then explained as a structural defect of the (M6) verticalization discipline in GPT-5.5's round-5 review.

Where Claude was weaker than GPT-5.5. Concretely on this project: (i) Claude did not spot architectural defects in its own drafts. The round-6 critical bug $\mathrm{lab}(u, v) := L_Q(q(u), q(v))$ with $L_Q(\pi, \pi) = \mathsf{EQ}$ collapsing distinct blocking laps into semantic equality was a basic invariant violation of the central blocking-not-equality slogan; GPT-5.5 caught it on review. (ii) Patch-style fixes accumulated complexity. Claude's response to a named gap was usually to add machinery (recursive verticalization, port-indexed mate clusters, declared cross-label functions, modal-depth recurrences, the validity-clause numbering growing from (V1)–(V10) to (V1)–(V15)). Each patch closed the named gap but introduced new internal-consistency issues. GPT-5.5's round-7 fix changed the indexing domain instead of adding a layer: five round-5 / round-6 validity clauses collapsed into one (V6) triple-shape clause, and the manuscript dropped from 39 pages (round-6 consolidation) to 14 pages (round-7 sketch). (iii) Claude did not reliably know when to stop iterating.

Where Claude held its own. Implementation work—translating an architectural idea into running Python, debugging against an independent reasoner, building cross-validation harnesses, keeping the tests green across thousands of generated concepts—is closer to ordinary programming, at which current models are strong. Documentation and audit-trail discipline (per-round superseded-paper markers, the OUTDATED archive, self-contained verification scripts) favoured patient editorial labour over insight. Adversarial review of someone else's code, when the reviewer started in a fresh session with no context, was effective: Claude Opus 4.7 identified the twelve-counterexample family in the cover-tree implementation that forced the (CT5) role-path compatibility addition (Remark 6.6). And steering the cross-validation pipeline—knowing which stress tests would discriminate between candidate proofs—requires understanding both the proof and how to falsify it.

What both models did well. Long-context coherence across 30+ page proofs and many editing rounds was qualitatively different from the previous LLM generation. Tool use (Python, LaTeX, git, file management, search) was routine. Seven-round adversarial dialogue between Claude and GPT-5.5—each model reviewing the other's drafts, producing counterexamples, naming defects, proposing fixes—worked as substantive technical exchange. Hallucination was not the dominant failure mode in this project; over-elaboration was.

What both models did poorly. Original mathematical insight was the human contribution. Without Wessel's whiteboard sketch (Figure 1) the project would not have had a target architecture to formalise. Both models also made architectural errors in their own drafts that the other caught; neither caught them in self-review. And neither reliably calibrated the cost of a fix: both round-4 (M6) verticalization and round-5/6 (V11)–(V15) were locally reasonable responses that turned out, on review, to be over-elaborated.

What this means in practice. Frontier LLMs in May 2026 can take a non-trivial open problem in mathematical logic and, over weeks of iteration with a human director and adversarial cross-review between models, produce a candidate proof together with executable artefacts that the community can verify, refute, or improve. They do not replace human expert review, and they do not generate the original intuition. But the proof that arrived at round 7 is qualitatively different from what any one of the three participants would have produced alone, and the cross-validation campaign that surrounds it is qualitatively different from what would have been feasible without the implementation speed and tireless test-writing the LLMs brought to the project.

This is a discussion piece, not a settled result. The work has not been peer-reviewed by human experts. The proof may still contain defects that another round of adversarial review would surface. But the meta-research point—*that this kind of work is now possible, and that the result is worth engaging with on its technical merits*—is the more important contribution. We invite the description logic community to engage with both layers.

8.4. A note on citation verification

Because AI systems are known to hallucinate bibliographic references, all external citations in this paper and its companion papers were verified via web search against journal landing pages, CEUR-WS/LIPIcs/arXiv metadata, and indexed proceedings. The audit caught one fully hallucinated reference (the non-existent "Rybina–Zakharyaschev 2001"—the genuine paper at that venue is Reynolds and Zakharyaschev, *On the products of linear modal logics*, JLC 11(6), 2001) and several transcription errors (incorrect coauthors, page ranges, venues, and journal names). All fixes are documented in the project's `CONVERSATION.md` log with commit hashes. Citations in this paper have been checked, but readers should still verify any reference before citing it.

Declaration on Generative AI

In preparing this work, the author (Wessel) made significant use of generative-AI language tools: **Claude** (Anthropic, Opus 4.7), **GPT-5.4 Pro** (OpenAI), and **GPT-5.5** (OpenAI). Specifically, the AI tools were used to:

- formalise the mathematical proofs (split-forest semantics, completeness extraction across seven review rounds, the round-7 pair-shape / incidence-tag architecture);
- implement the cover-tree tableau decision procedure (≈350 lines of Python), the quasimodel reasoners, and the WP1–WP9 verification scripts;
- conduct computational verification campaigns (911-concept cross-validation, 775-concept decomposition test, the round-7 stress-concept verifications);
- draft the LaTeX manuscript text, including this overview and all companion papers in the repository.

The division of labour between the human author and the AI tools is documented in detail in §8.3 of this paper.

Responsibility. By signing this work as its author, Wessel takes full responsibility for all of its contents, irrespective of how the contents were generated. Where the generative-AI tools produced inappropriate language, plagiarised content, biased content, errors, mistakes, incorrect references, or misleading content that was included in the work, that responsibility rests with the author, not with the tools.

Authorship. Per arXiv's content moderation policy, generative-AI tools are not listed as authors of this paper. The AI tools are disclosed here as the substantial methodological instruments they were.

Direction. The research direction—problem formulation, the key conceptual insights (cover-tree intuition, PPI-tree orientation, EQ-splitting, DR-rigidity), the correction of architectural errors (notably GPT-5.4's initial inclusion of PP/PPI as cross-edges, which would have defeated the construction), and the high-level judgement calls about when to converge—was provided by Wessel; the AI tools performed the detailed formalisation, implementation, verification, and manuscript drafting under that direction.

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